

Natnael Alemseged Astaw

AI Agent & Forward-Deployed Engineer | Senior Full-Stack Developer

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Upwork](#)

PROFILE

Software engineer with 4+ years of experience shipping production AI, web, and mobile products. Builds agentic systems, RAG, evaluation and fine-tuning pipelines, LLM-powered APIs, and polished cross-platform applications using Python, TypeScript, Flutter, and AWS. Experienced in taking ambiguous product requirements from discovery through deployment, observability, and iteration.

EXPERIENCE

Senior Software Engineer

Jul 2025 - Present

Hire Armada | United States (Remote)

- Engineer AI-driven backends and product features with Node.js, FastAPI, Next.js, Flutter, Prisma, LangGraph, MCP, Composio, and vector memory.
- Built a social discovery platform backend supporting 10K+ daily active users, real-time matching and messaging, and 99.9% uptime.
- Shipped a personalized affirmation product with Gemini chat, home-screen widgets, and adaptive feeds, increasing retention by 35%.
- Delivered an agentic collaboration workspace with real-time events, email automation, semantic search, and tagged agents, improving team productivity by 20%.

Senior Full-Stack Developer

Jan 2025 - May 2026

Startup Agile | United States (Remote)

- Led Next.js, React, React Native, and Expo delivery across web and mobile products; completed milestones 15% ahead of schedule.
- Implemented OTA updates, deep linking, Firebase notifications, Google Maps, AI assistants, and product analytics while reducing mobile load time by 35%.
- Improved Next.js rendering, SEO, and Core Web Vitals, contributing to 50%+ web traffic growth and a 30% performance improvement.
- Standardized TypeScript, tRPC, Prisma, Playwright, and Detox workflows, reducing production defects by 65%.

AI Engineer

Jan 2025 - Aug 2025

DataCore Software | United States (Remote, Contract)

- Built FastAPI services for real-time detection, recognition, transcription, summarization, and translation using self-hosted LLMs.
- Deployed containerized inference workloads to AWS EKS with sub-200 ms inference and 99.9% uptime at peak load.
- Fine-tuned open models and optimized GPU inference with TensorRT and ECR images, reducing cost by 12%; automated delivery with Terraform and GitHub Actions.

Senior Mobile Developer

Aug 2024 - Sep 2025

Qemer Software Technology PLC | Addis Ababa, Ethiopia (On-site)

- Led delivery of 10+ Flutter and Ionic applications, accelerating release cycles by 40%.
- Improved application architecture and runtime performance, cutting load times by up to 50%.
- Mentored junior engineers and established reusable delivery patterns, reducing onboarding time by 30%.

EARLIER EXPERIENCE

- ERP Functional Consultant

Red Cloud ICT Solutions | Addis Ababa, Ethiopia

Mar 2023 - Sep 2024

- Progressed from Junior Consultant to Functional Consultant Assistant, supporting ERP discovery, configuration, testing, user enablement, and implementation delivery.
- Freelance Web Developer

Independent | Remote

Aug 2022 - Mar 2023

- Designed, built, and delivered responsive web solutions directly with clients from requirements through launch.

SELECTED INDEPENDENT DELIVERY

- Social Music Party Game:

Delivered a complete client product engagement from Jan-Jun 2026; received a verified 5.0 Upwork rating.
- AI Chatbot Engineering:

Completed a focused AI chatbot engagement in Jan 2026; received a verified 5.0 Upwork rating.
- Flutter Inventory Tracker:

Shipped and supported a cross-platform inventory application in 2025; received a verified 5.0 Upwork rating.
- Upwork Top Rated

| 100% Job Success

| Verified 5-star client outcomes

TECHNICAL SKILLS

- AI & Agents:

LangGraph, RAG, agent orchestration, LLM-as-a-judge, statistical evaluation, fine-tuning, LoRA, DPO/SimPO, structured generation, zero-shot prompting, MCP, vector databases
- ML Experimentation:

Jupyter Notebook, Google Colab, reproducible experiments, benchmark design, error analysis, sampling and calibration analysis
- Backend & Data:

Python, FastAPI, Node.js, Express.js, TypeScript, PostgreSQL, MongoDB, Redis, Supabase, Firebase, Prisma, GraphQL, REST APIs, Socket.io
- Web & Mobile:

Next.js, React, React Native, Expo, Flutter, Dart, Ionic, Redux Toolkit, TanStack Query, BLoC
- Cloud & Delivery:

AWS, Docker, Kubernetes/EKS, Terraform, GitHub Actions, Prometheus, Grafana, CI/CD, microservices, n8n

SELECTED TECHNICAL WRITING

- When Generic Benchmarks Fail: Building a Sales-Domain Evaluation Bench from Scratch

DEV Community | May 2026

Why Pairing Your Bootstrap Is Necessary. And When It Stops Helping

DEV Community | May 2026

CERTIFICATIONS

- AI Agent Engineering, AI Evaluation & Forward-Deployed Engineering

10 Academy | Issued May 2026

AWS Certified Cloud Practitioner

AWS | Oct 2023 - Feb 2027

CCNA: Introduction to Networks

Cisco | Issued Nov 2021
- AWS Certified Solutions Architect - Associate

AWS | Feb 2024 - Feb 2027

Founders Academy

ALX Africa | Issued Jul 2024

EDUCATION

- Bachelor of Science in Information Systems

Hawassa University | GPA: 3.57

Jan 2018 - Dec 2022